WTW158 – Memo Sem test 1 2012

SECTION A: MULTIPLE CHOICE QUESTIONS - 15 MARKS

READ THE FOLLOWING INSTRUCTIONS

1. YOU HAVE TO START WITH SECTION A, AS YOU HAVE TO HAND IN THE OPTIC READER FORM AFTER 60 MINUTES.
2. Use a soft pencil.
3. Use **SIDE 2** of the optic reader form. **(If side 1 is used it can not be marked.)**
   Fill in your personal information in pencil on side 2.
   Fill in your student number from top to bottom and then code it.
   If you make a mistake when coding your student number or if the student number is not filled in, you will get no marks for section A.
   Answer questions 1 to 15 on the optic reader form on **SIDE 2**.
4. As you may not erase wrong answers on the optic reader form, first circle your answers on this paper and only mark the answers on the form when you are certain of your choice.
5. Do all scribbling on the facing page. It will not be marked.
6. **You may start with section B as soon as you have marked your answers on the optic reader form.**

AFDELING A : MEERVOUDIGE KEUSEVRAE - 15 PUNTE

LEES DIE VOLGENDE INSTRUKSIES

1. U MOET MET AFDELING A BEGIN WANT U MOET DIE MERKLEESVORM NA 60 MINUTE INHANDIG.
2. Gebruik 'n sagte potlood.
3. Gebruik **KANT 2** van die merkleesvorm.
   (Indien kant 1 gebruik word kan dit nie nagesien word nie).
   Vul u persoonlike inligting in potlood in op kant 2.
   Vul u studentenommer van bo na onder in en kodeer dit.
   Indien u studentenommer verkeerd gekodeer is of nie ingevul is nie, sal u nie punte vir afdeling A kry nie.
   Beantwoord vrae 1 tot 15 op die merkleesvorm se **KANT 2**
4. U mag nie verkeerde antwoorde uitvee op die merkleesvorm nie, omkring dus u antwoorde eers in hierdie vraestel en merk die antwoorde op die vorm as u seker is van u keuse.
5. Doen alle krapwerk op die teenblad. Dit word nie nagesien nie.
6. **U kan met afdeling B begin sodra u die antwoorde op die merkleesvorm ingevul het.**

Question 1 / Vraag 1

The solution set of $\frac{x-4}{-2} \leq 0$ is

Die oplossingsversameling van $\frac{x-4}{-2} \leq 0$ is

$\frac{x-4}{-2} \leq 0 \Rightarrow x-4 \geq 0$
$\Rightarrow x \geq 4$

[1a] ϕ (empty set / leë versameling)	[1b] $\mathbb{R}$	[1c] $(-\infty,4]$	[1d] $[4,\infty)$
[1e] None of these / Geen van hierdie			

Question 2 / Vraag 2 $-40 \le x \le 100 \Rightarrow 80 \ge -2x \ge -200 \Rightarrow 90 \ge 10-2x \ge -190$

The relationship between x and y is given by $y = 10 - 2x$.
What interval on the y-axis corresponds to the interval $x \in [-40, 100]$ on the x-axis?

Die verband tussen x en y word gegee deur $y = 10 - 2x$.
Watter interval op die y-as stem ooreen met die interval $x \in [-40, 100]$ op die x-as?

[2a] $x \in [90, 190]$	[2b] $x \in [-90, 190]$	[2c] $x \in [-190, 90]$	[2d] $x \in [-190, -90]$
[2e] None of these / Geen van hierdie			

Question 3 / Vraag 3

If $(f \circ g \circ h)(x) = \sqrt{e^{1-x}}$, $h(x) = 1 - x$ and $f(x) = \sqrt{x}$ then $g(x) =$

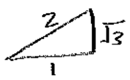
Indien $(f \circ g \circ h)(x) = \sqrt{e^{1-x}}$, $h(x) = 1 - x$ en $f(x) = \sqrt{x}$ dan is $g(x) =$

[3a] x	[3b] e^{1-x}	[3c] e^x	[3d] None of these / Geen van hierdie

$g(x) = e^x$
$(f \circ g \circ h)(x) = f \circ g(1-x)$
$= f(g(1-x))$
$= f(e^{1-x})$
$= \sqrt{e^{1-x}}$

Question 4 / Vraag 4

$\cos(\frac{4\pi}{3}) =$



[4a] $\frac{1}{2}$	[4b] $-\frac{1}{2}$	[4c] $\frac{\sqrt{3}}{2}$	[4d] $-\frac{\sqrt{3}}{2}$	[4e] $\frac{1}{\sqrt{2}}$	[4f] $-\frac{1}{\sqrt{2}}$
[4g] None of these / Geen van hierdie					

Question 5 / Vraag 5

$\tan^{-1}(-\sqrt{3}) =$ (Notation: $\tan^{-1} = \arctan$ / Notasie: $\tan^{-1} = \text{bgtan}$)

$y = \tan^{-1}(-\sqrt{3})$

[5a] $\frac{2\pi}{3}$	[5b] $\frac{5\pi}{3}$	[5c] $-\frac{\pi}{3}$	[5d] $\frac{5\pi}{6}$	[5e] $\frac{11\pi}{6}$	[5f] $-\frac{\pi}{6}$
[5g] None of these / Geen van hierdie					

$\Leftrightarrow \tan y = -\sqrt{3}$ and
$y \in (-\frac{\pi}{2}, \frac{\pi}{2})$

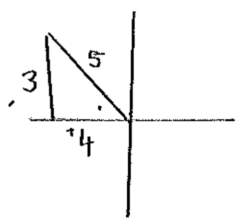
Question 6 / Vraag 6

If $\sin x = \frac{3}{5}$, $\frac{\pi}{2} < x < \pi$, then $\cot x =$

Indien $\sin x = \frac{3}{5}$, $\frac{\pi}{2} < x < \pi$, dan is $\cot x =$

[6a] $\frac{4}{3}$	[6b] $-\frac{4}{3}$	[6c] $\frac{5}{3}$	[6d] $-\frac{5}{3}$	[6e] $\frac{3}{5}$	[6f] $-\frac{3}{5}$
[6g] None of these / Geen van hierdie					

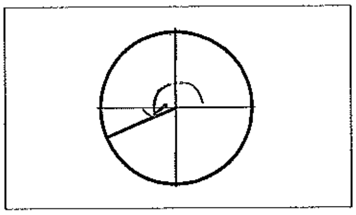
$\sin x = \frac{3}{5}$



$\cot x = \frac{1}{\tan x}$
$= \frac{\cos x}{\sin x} = -\frac{4}{5} \times \frac{5}{3}$
$= -\frac{4}{3}$

Question 7 / Vraag 7

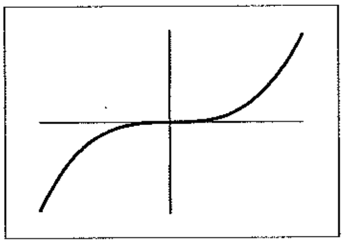
The angle in the sketch is / Die hoek in die skets is



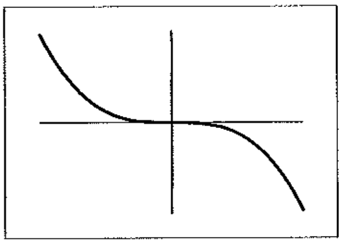
[7a] $\frac{15\pi}{8}$	[7b] $\frac{9\pi}{8}$	[7c] $\frac{7\pi}{8}$	[7d] $\frac{\pi}{8}$

Use the six graphs below to answer Questions 8 and 9:

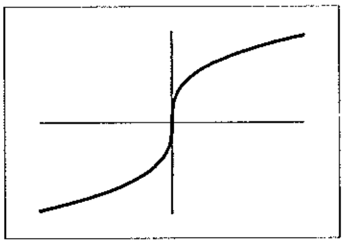
Gebruik die ses grafieke hieronder om Vrae 8 en 9 te beantwoord:



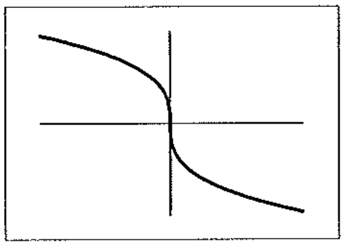
I



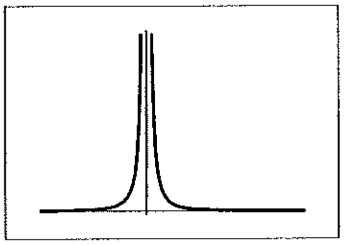
II



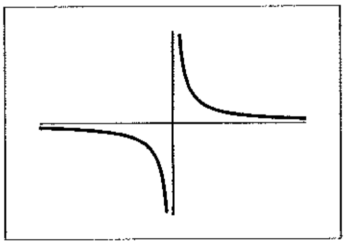
III



IV



V



VI

Question 8 / Vraag 8

The graph of $y = \frac{1}{x^8}$ is number / Die grafiek van $y = \frac{1}{x^8}$ is nommer

[8a] **I**	[8b] **II**	[8c] **III**	[8d] **IV**	[8e] **V**	[8f] **VI**

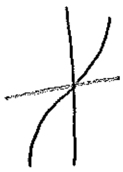
Similar shape as $y = \frac{1}{x^2}$

Question 9 / Vraag 9

The graph of $y = -x^7$ is number / Die grafiek van $y = -x^7$ is nommer

[9a] **I**	[9b] **II**	[9c] **III**	[9d] **IV**	[9e] **V**	[9f] **VI**

Similar shape as $-x^3$



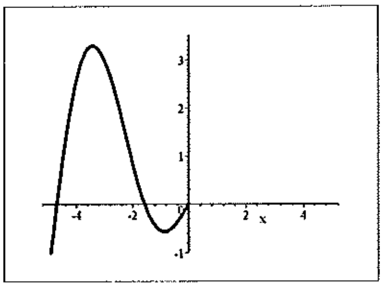
$y = x^3$

$y = -x^3$

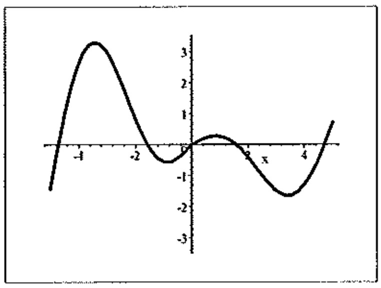
Question 10 / Vraag 10

A function f has domain $[-5,5]$ and a portion of its graph is shown in Sketch **I** below.
In Sketch **II** below the graph of f is completed and it can be seen that f is an

'n Funksie f het definisieversameling $[-5,5]$ en 'n deel van die grafiek word in Skets **I** hieronder gegee.
In Skets **II** hieronder is die grafiek van f voltooi en dit kan gesien word dat f 'n



I



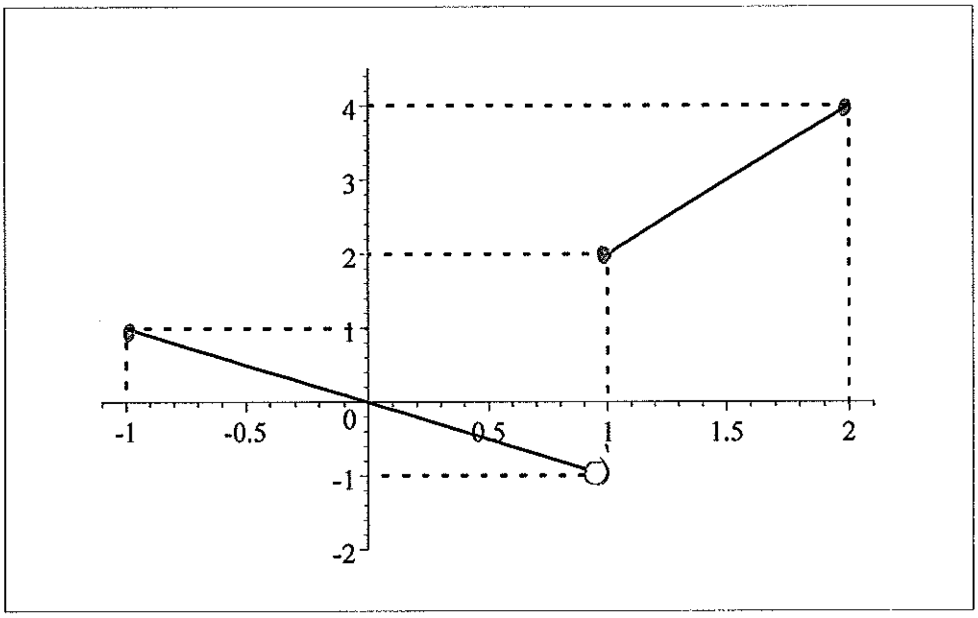
II

[10a] even function / ewe funksie is	[10b] odd function / 'n onewe funksie is
[10c] None of these / Geen van hierdie	

Question 11 / Vraag 11

Consider the graph of a one-to-one function f below.
The domain of the inverse function, f^{-1}, is

Beskou die grafiek van 'n een-tot-een funksie f hieronder.
Die definisieversameling van die inverse funksie, f^{-1}, is

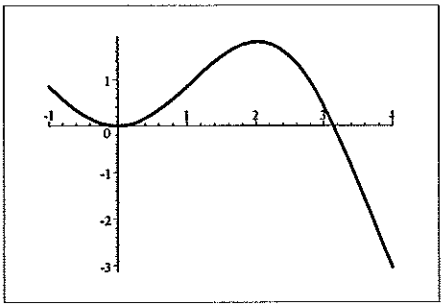


[11a] $[-1,2]$	[11b] $[-1,4]$	[11c] $[-1,1]\cup[2,4]$	[11d] $(-1,1]\cup[2,4]$
[11e] None of these / Geen van hierdie			

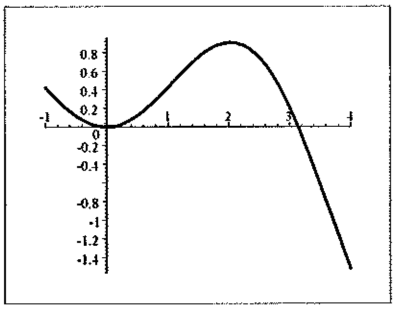
Domain of f^{-1} = range f = $(-1,1]\cup[2,4]$

Use the three graphs below to answer Questions 12 and 13:

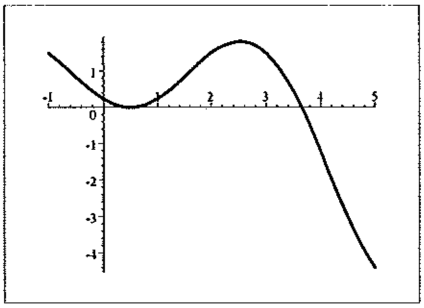
Gebruik die drie grafieke hieronder om Vrae 12 en 13 te beantwoord:



$y = f(x)$



$y = g(x)$



$y = h(x)$

Question 12 / Vraag 12

$g(x) =$

[12a] $f(x-0.5)$	[12b] $f(x+0.5)$	[12c] $0.5f(x)$	[12d] $\frac{f(x)}{0.5}$

Question 13 / Vraag 13

$h(x) =$

[13a] $f(x-0.5)$	[13b] $f(x+0.5)$	[13c] $0.5f(x)$	[13d] $\frac{f(x)}{0.5}$

Question 14 / Vraag 14

Consider the three equations below:

Beskou die drie vergelykings hieronder:

I $|x+1| = -3$ **II** $x^2 + x = -3$ **III** $\operatorname{cosec} x = -3$

$\Rightarrow \sin x = -\frac{1}{3}$ has solutions

I $|x+1| = -3$ has no solutions as $|a| \geq 0$ for all a.

II $x^2 + x + 3 = 0$

$\Delta = b^2 - 4ac = 1 - 4 \times 1 \times 3$
$= -11 < 0$

no solution

Which of the equations has(have) no solution?

Watter van die vergelykings het nie 'n oplossing nie?

[14a] only / slegs I	[14b] only / slegs II	[14c] only / slegs I and / en II
[14d] only / slegs III	[14e] only / slegs I and / en III	[14f] only / slegs II and / en III
[14g] None of these / Geen van hierdie		

Question 15 / Vraag 15

The solution set of $|x^2 - 1| = 15,\ x \in \mathbb{R}$ is

Die oplossingsversameling van $|x^2 - 1| = 15,\ x \in \mathbb{R}$ is

[15a] $\{16\}$	[15b] $\{4, -4, \sqrt{-14}, -\sqrt{-14}\}$	[15c] $\{4, -4\}$
[15d] None of these / Geen van hierdie		

[15]

$x^2 - 1 = 15$ or $x^2 - 1 = -15$

$\Rightarrow x^2 = 16$ or $x^2 = -14$

$\downarrow$ no real solution

$\Rightarrow x = \pm 4$

SECTION B: 33 MARKS

READ THE FOLLOWING INSTRUCTIONS

1. Answer all the following questions on this paper in ink.
2. Work done in pencil or in red ink will not be marked.
3. You are not allowed to use correction fluid ("Tipp-Ex").
4. Do all scribbling on the facing page. It will not be marked.
   If you need more space for an answer, use the facing page and indicate it clearly **by framing it**.
5. Show all steps and calculations and explain your answers.

AFDELING B: 33 PUNTE

LEES DIE VOLGENDE INSTRUKSIES

1. Beantwoord die volgende vrae op hierdie vraestel in ink.
2. Werk wat in potlood of rooi ink beantwoord, is word nie nagesien nie.
3. U mag nie korrigeer-vloeistof ("Tipp-Ex") gebruik nie.
4. Doen alle krapwerk op die teenblad. Dit word nie nagesien nie.
   As u nog ruimte vir 'n antwoord nodig het, gebruik die teenblad en dui dit duidelik aan **deur 'n raam daarom te trek.**
5. Toon alle stappe en berekeninge en verduidelik u antwoorde.

Question 16 / Vraag 16

i Use a suitable theorem about the absolute value to solve for x if $|3x+2| \geq 4$.
Write the answer in terms of intervals.

Gebruik 'n geskikte stelling oor absolute waarde om vir x op te los as $|3x+2| \geq 4$.
Skryf die antwoord in terme van intervalle.

$|3x+2| \geq 4$

$\Rightarrow 3x+2 \geq 4$ or $3x+2 \leq -4$

$\Rightarrow 3x \geq 2$ or $3x \leq -6$

$\Rightarrow x \geq 2/3$ or $x \leq -2 \Rightarrow x \in (-\infty,-2] \cup [2/3,\infty)$

[2]

ii Solve for x if $\frac{2-x}{|x-5|} < 0$.

Los op vir x indien $\frac{2-x}{|x-5|} < 0$.

$\frac{2-x}{|x-5|} < 0 \Rightarrow 2-x < 0$ and $|x-5| \neq 0$ because $|x-5| \geq 0$

$\Rightarrow x > 2$ and $x \neq 5$

$\Rightarrow x \in (2,5) \cup (5,\infty)$

[2]

Question 17 / Vraag 17

Solve for x : / Los op vir x :

i $3\ln x - \ln x^2 < 5$

$\Rightarrow 3\ln x - 2\ln x < 5$

$\Rightarrow \ln x < 5$

$\Rightarrow x < e^5$ and $x > 0$ (because domain of $\ln x$ is $(0, \infty)$)

$\Rightarrow 0 < x < e^5 \Rightarrow x \in (0, e^5)$

OR $3\ln x - \ln x^2 < 5$

$\Rightarrow \ln x^3 - \ln x^2 < 5$

$\Rightarrow \ln \frac{x^3}{x^2} < 5$

$\Rightarrow \ln x < 5$

[3]

ii $\frac{e^{x^2} e^x}{e^2} = 1$

$\Rightarrow e^{x^2+x-2} = 1 \Rightarrow \ln e^{x^2+x-2} = \ln 1$

$\Rightarrow x^2 + x - 2 = 0$

$\Rightarrow (x+2)(x-1) = 0$

$\Rightarrow x = -2$ or $x = 1$

[3]

Question 18 / Vraag 18

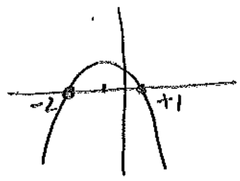
Find the domain of the function $f(x) = \sqrt{2 - x - x^2}$. Show ALL the steps.

Bepaal die definisieversameling van die funksie $f(x) = \sqrt{2 - x - x^2}$. Toon ALLE stappe.

$2 - x - x^2 \geq 0 \Rightarrow (2+x)(1-x) \geq 0$

$\Rightarrow x \in [-2, 1]$

using graph $\longrightarrow$



$\therefore$ Domain $= \{x \mid -2 \leq x \leq 1\} = [-2, 1]$

[3]

Question 19 / Vraag 19

Solve for x if $2\sin^2 x + \sin x - 1 = 0$.

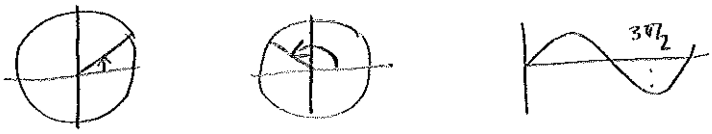
Los op vir x as $2\sin^2 x + \sin x - 1 = 0$.

$2\sin^2 x + \sin x - 1 = (2\sin x - 1)(\sin x + 1) = 0$

$\Rightarrow \sin x = \frac{1}{2}$ or $\sin x = -1$

$\Rightarrow x = \frac{\pi}{6} + 2k\pi$ or $x = \frac{5\pi}{6} + 2k\pi$ or $x = \frac{3\pi}{2} + 2k\pi$

$k \in \mathbb{Z}$

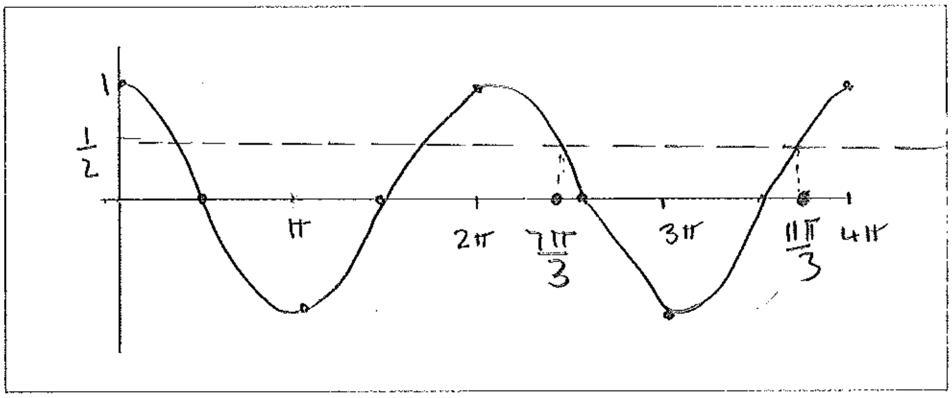


[4]

Question 20 / Vraag 20

Sketch the graph of $y = \cos x$, $x \in [0, 4\pi]$.

Skets die grafiek van $y = \cos x$, $x \in [0, 4\pi]$.



Solve for x if $\cos x - \frac{1}{2} > 0$, $x \in [2\pi, 4\pi]$.
Indicate your answer on the sketch above.

Los op vir x indien $\cos x - \frac{1}{2} > 0$, $x \in [2\pi, 4\pi]$.
Dui u antwoord op die skets hierbo aan.

$\cos x = \frac{1}{2} \Rightarrow x = \frac{\pi}{3} + 2k\pi$ or $x = \frac{5\pi}{3} + 2k\pi$, $k \in \mathbb{Z}$

$\cos x = \frac{1}{2}$, $x \in [2\pi, 4\pi] \Rightarrow x = \frac{\pi}{3} + 2\pi = \frac{7\pi}{3}$

or $x = \frac{5\pi}{3} + 2\pi = \frac{11\pi}{3}$

[4]

$\cos x - \frac{1}{2} > 0 \Rightarrow \cos x > \frac{1}{2} \Rightarrow x \in \left[2\pi, \frac{7\pi}{3}\right) \cup \left(\frac{11\pi}{3}, 4\pi\right]$

Question 21 / Vraag 21

Consider the function $f(x) = x|x-1|$.

Beskou die funksie $f(x) = x|x-1|$.

i Calculate $f(-3)$. / Bepaal $f(-3)$.

$$f(-3) = -3 \times |-4| = -3 \times 4 = -12$$

[1]

ii Write the function $y = f(x)$ as a piecewise defined function.

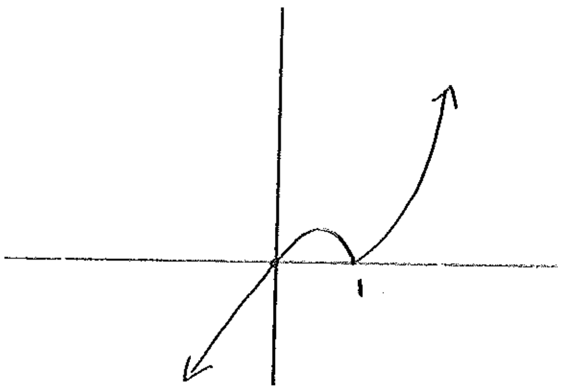
Skryf die funksie $y = f(x)$ as 'n stukgewys-gedefinieerde funksie.

$$f(x) = x|x-1| = \begin{cases} x(x-1) & \text{if } x-1 \geq 0 \\ -x(x-1) & \text{if } x-1 < 0 \end{cases}$$

$$= \begin{cases} x(x-1) & \text{if } x \geq 1 \\ -x(x-1) & \text{if } x < 1 \end{cases}$$

[2]

iii Sketch the graph of $y = f(x)$ / Skets die grafiek van $f(x)$.



[2]

Question 22 / Vraag 22

i Write down the definition of the function $\cos^{-1}x = \arccos x$.

Gee die definisie van die funksie $\cos^{-1}x = \text{bgcos}\, x$.

$y = \cos^{-1}x \iff \cos y = x$ and $y \in [0, \pi]$

[1]

ii Complete the following cancellation property:

Voltooi die volgende kansellasie eienskap:

$(\cos(\cos^{-1}x)) = x$ for / vir $x \in [-1, 1]$

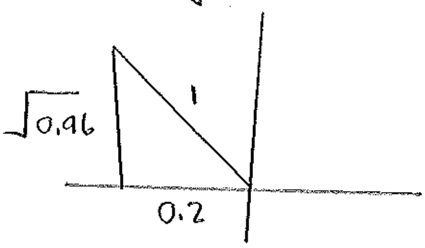
[1]

iii Calculate $\tan(\cos^{-1}(-0.2))$. / Bepaal $\tan(\cos^{-1}(-0.2))$.

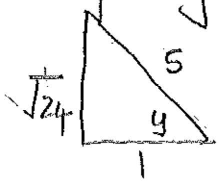
Let $y = \cos^{-1}(-0.2)$

$\Rightarrow \cos y = -0.2$ and $y \in [0, \pi]$

[2]



or using $\cos y = -\frac{2}{10} = \frac{-1}{5}$



$\tan(\cos^{-1}(-0.2))$

$= \tan y$

$= -\frac{\sqrt{0.96}}{0.2}$

or $\tan y = \frac{\sqrt{24}}{1} = \sqrt{24}$

Question 23 / Vraag 23

i Write down the definition of an odd function.
Gee die definisie van 'n onewe funksie.

f is an odd function if $f(-x) = -f(x)$
for every x in the domain of f

[1]

ii Assume that $y = f(x)$ and $y = g(x)$ are odd functions with the same domains.
Is the function $y = (fg)(x)$ an odd function? Explain your answer.

Aanvaar dat $y = f(x)$ en $y = g(x)$ onewe funksies is met dieselfde definisieversamelings.
Is die funksie $y = (fg)(x)$ 'n onewe funksie? Verduidelik u antwoord.

$$(fg)(-x) = f(-x)\,g(-x) \quad \text{def of } fg$$
$$= -f(x) \times -g(x) \quad f \text{ and } g \text{ are odd}$$
$$= f(x)\,g(x)$$
$$= (fg)(x) \quad \text{def of } fg$$

[2]

$\therefore$ fg is an even function
fg is <u>not</u> an odd function